POUNAMU ELECTRONICS
RE_SWC
USER GUIDE

Rev 2

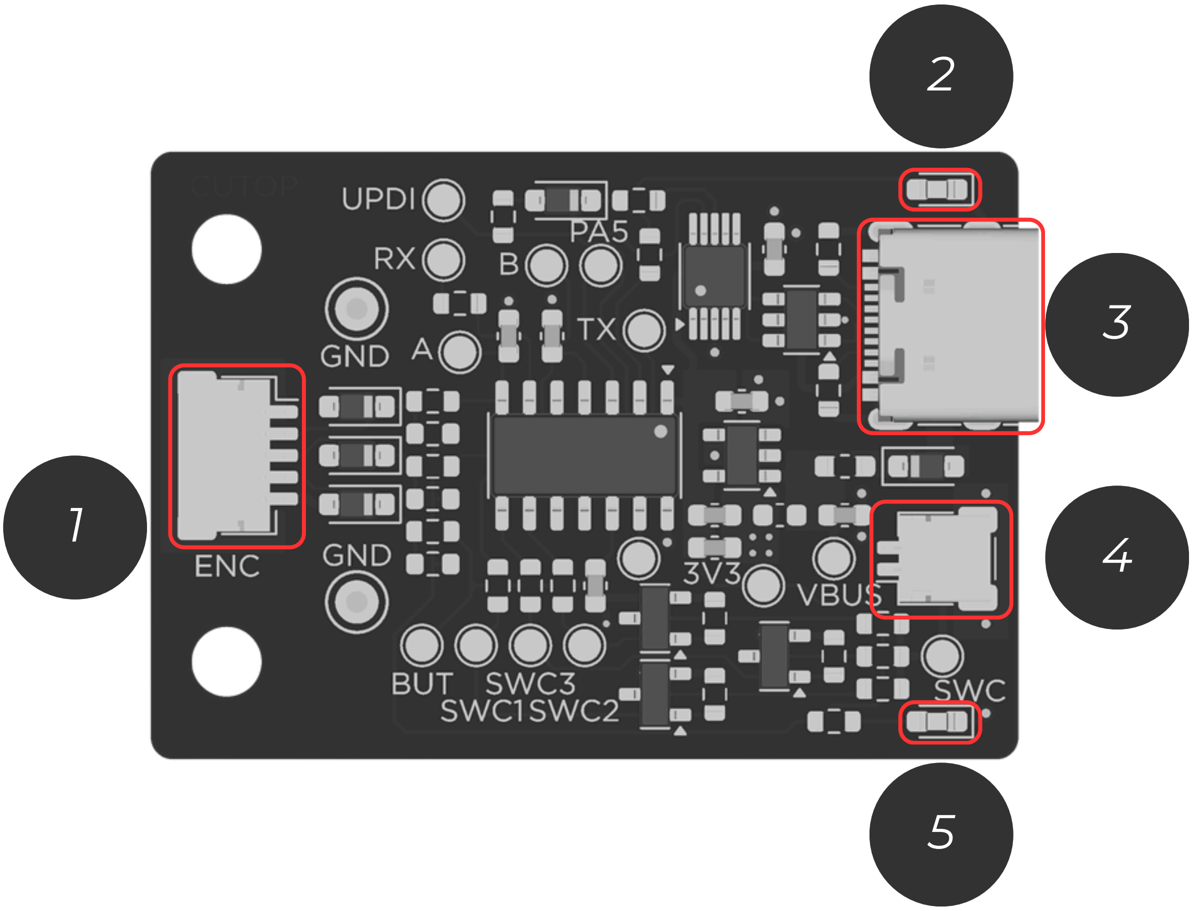
PE

POUNAMU ELECTRONICS

Contents

OVERVIEW

CONTROLLER BOARD



1. VOLUME KNOB CONNECTOR

2. WHITE STATUS LIGHT

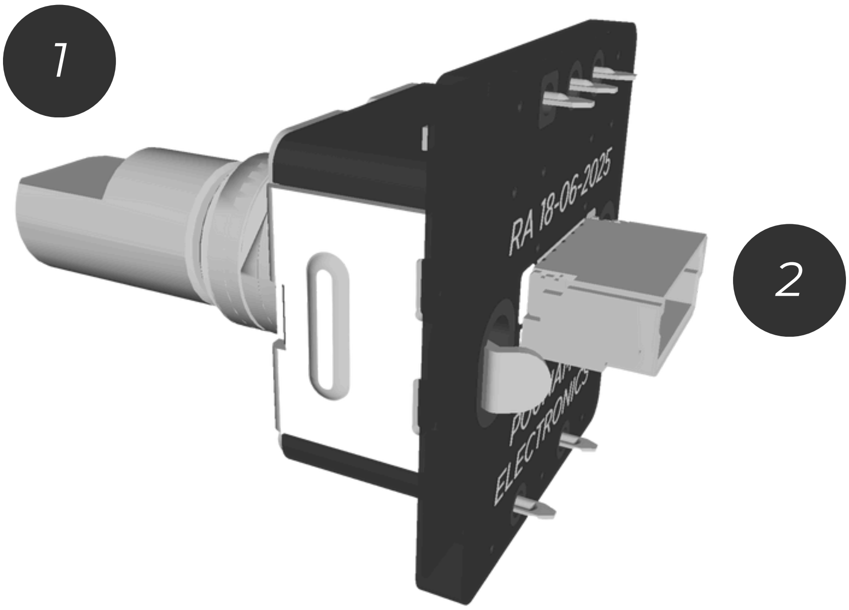
3. USB DATA & POWER IN

4. STEERING WHEEL CONTROL OUTPUT

5. RED POWER LIGHT

OVERVIEW

VOLUME KNOB BOARD

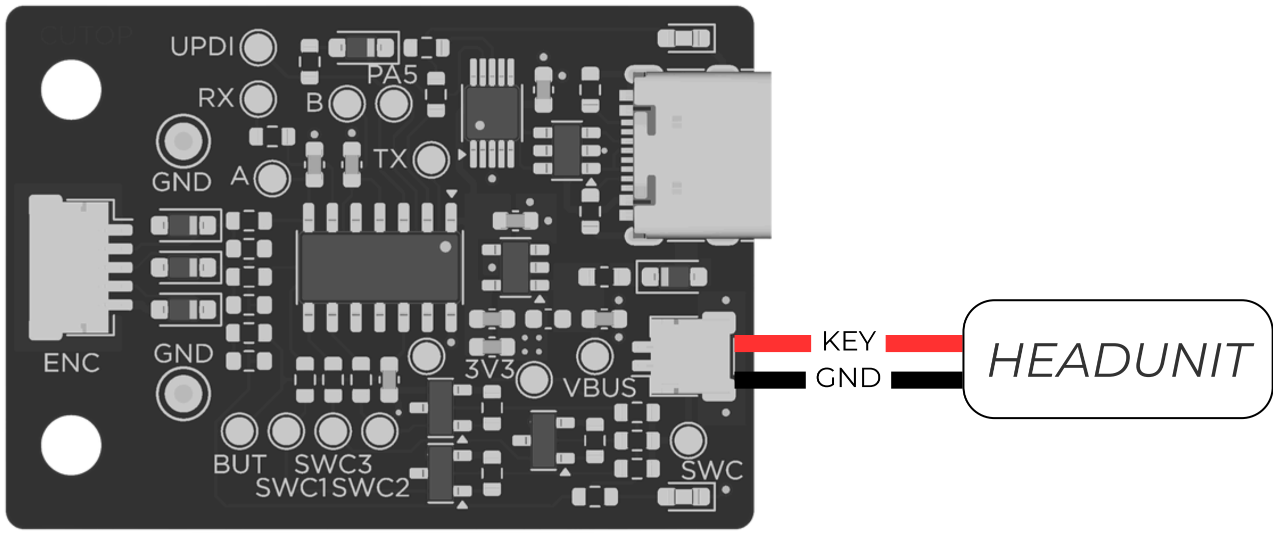


1. VOLUME KNOB

2. VOLUME KNOB CONNECTOR

ASSEMBLY
RESISTIVE LEARNING OUTPUT

- *Solder RE_SWC steering wheel control output to headunit steering wheel control input*
    - *RED wire to headunit steering wheel control input (KEY)*
    - *BLACK wire to headunit ground*
- *Plug the connector into RE_SWC*

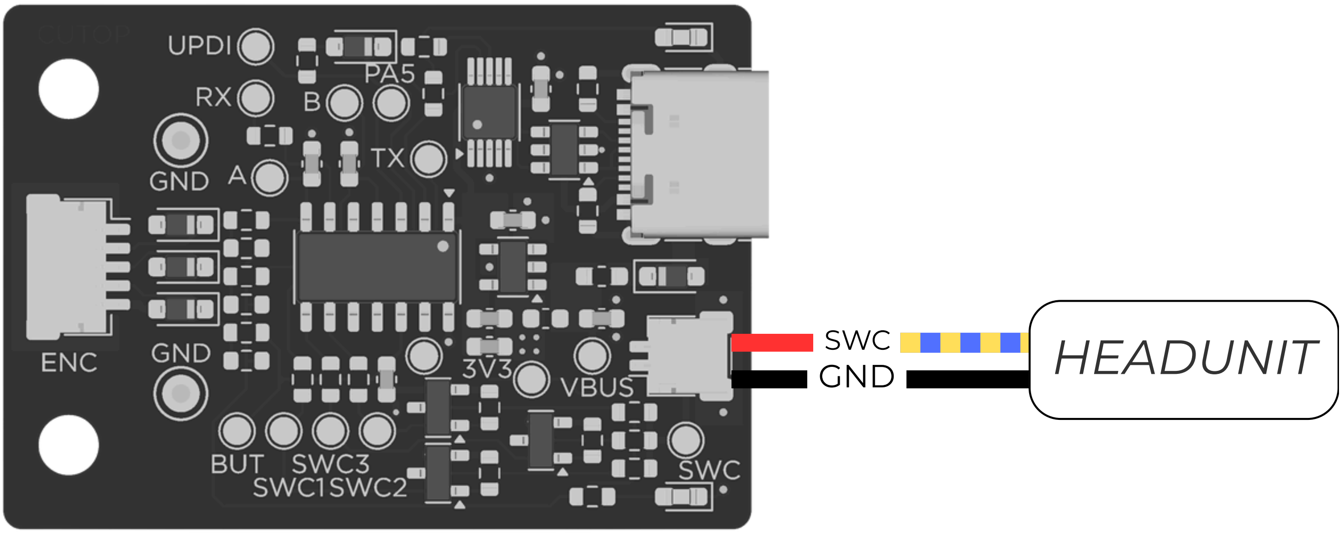


ASSEMBLY

JVC HEADUNIT SWC INTERFACE

Note: Only use this configuration if your JVC headunit does not support resistive SWC input

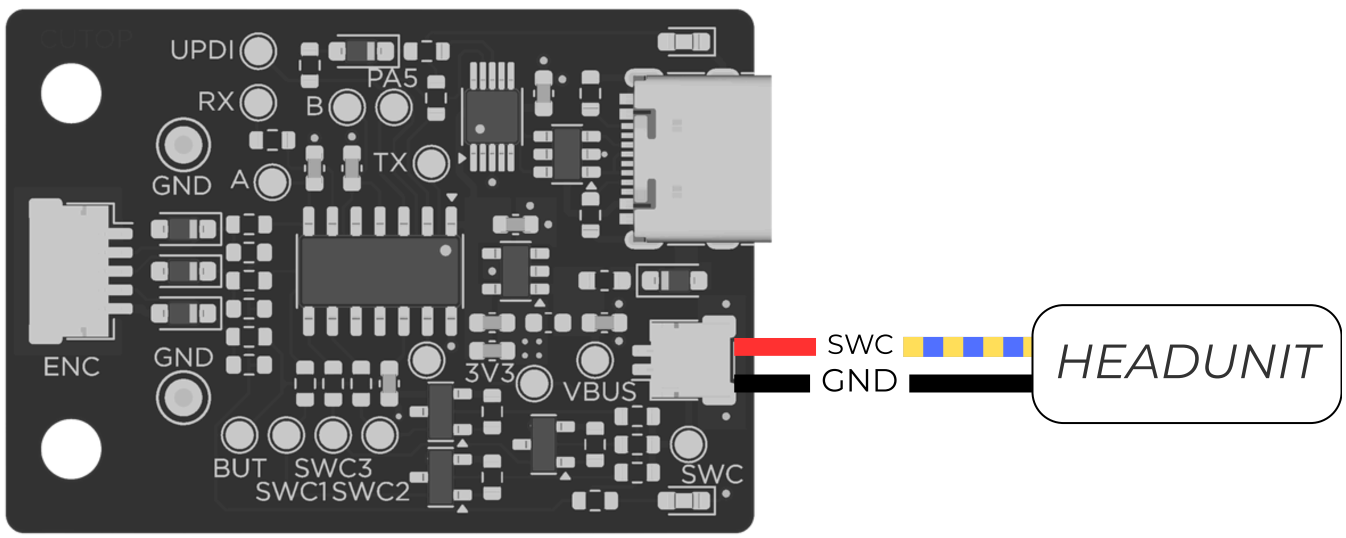
- *Solder RE_SWC steering wheel control output to headunit steering wheel control input*
  - *RED wire from RE_SWC to headunit steering wheel control input (BLUE with YELLOW trace)*
  - *BLACK wire from RE_SWC to headunit ground*
- *Plug the connector into RE_SWC*



ASSEMBLY
KENWOOD HEADUNIT SWC INTERFACE

Note: Only use this configuration if your Kenwood headunit does not support resistive SWC input

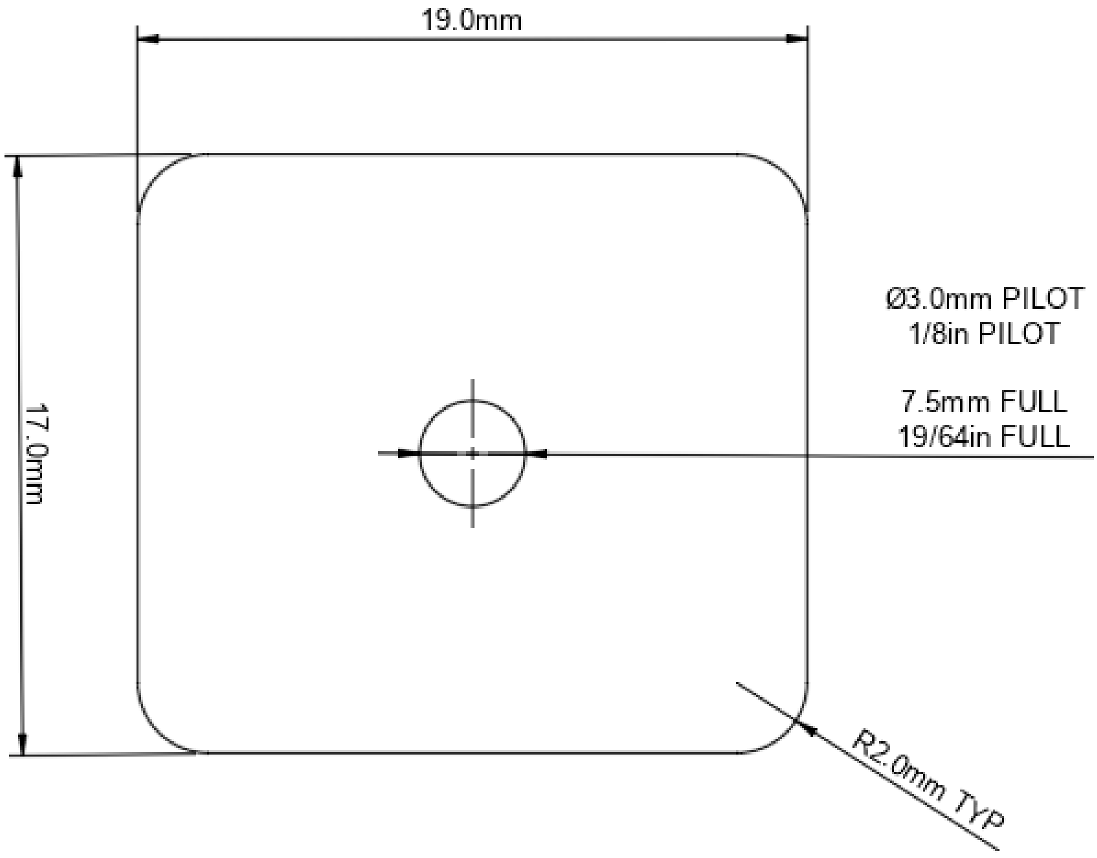
- *Solder RE_SWC steering wheel control output to headunit steering wheel control input*
    - *RED wire from RE_SWC to headunit steering wheel control input (BLUE with YELLOW trace)*
    - *BLACK wire from RE_SWC to headunit ground*
- *Plug the connector into RE_SWC*



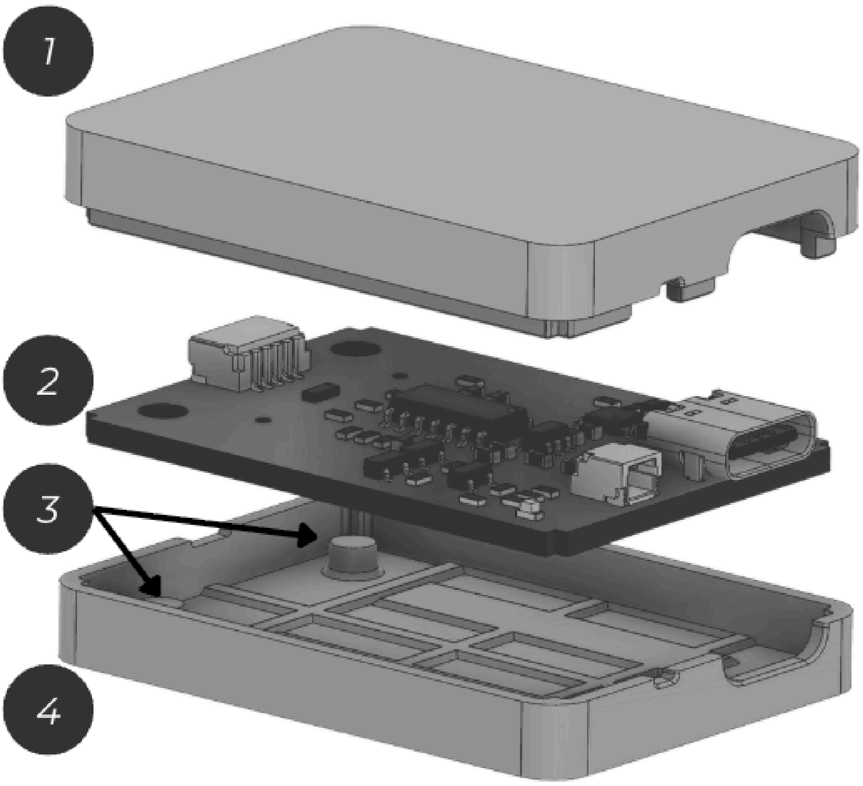
ASSEMBLY

VOLUME KNOB CONNECTION

- *Locate an area to fit your volume knob. Use the drilling template as a guide for clearance*
- *Use the drilling template included in the box or shown below to create a 3mm (1/8in) pilot hole for the volume knob*
- *Open the pilot hole with a 7.5mm (19/64in) drill bit*
- *Plug the 5-pin connector into the volume knob board*
- *Mount the volume knob to your panel. Secure it with the washer and nut included in the kit*
- *Plug the 5-pin volume knob connector into RE_SWC*



ASSEMBLY
CASE



1. CASE LID **2. SWC BOARD**

3. GUIDE POSTS **4. CASE BASE**

- *Ensure the volume knob and SWC output connectors are connected to the controller before proceeding*
- *Lower the SWC board into the case base using the guide posts for alignment and press into place. The volume knob and SWC output connectors will sit flush to the inside edge of the case*
- *Press the case lid into the case base. It should be a tight, snap fit. The connectors will be captive inside the case, unable to be pulled out*

ASSEMBLY
POWER

- *Plug the included USB-C cable into the RE_SWC*
- *When the RE_SWC is powered, the RED power light will illuminate*
- *When working correctly, rotating the volume knob or pressing the volume knob button will result in the WHITE status light illuminating briefly*
- *If the WHITE status light does not illuminate, ensure the volume knob connector is fully seated into both the controller and volume knob boards*

CONFIGURATION
HEADUNIT SELECTION

Note: This feature is only supported on devices with firmware version 2.0.0 and higher

- *Disconnect power from RE_SWC*
- *Press and hold the volume knob. Connect power to the RE_SWC while keeping the volume knob held*
- *Release the volume knob when the WHITE status light starts flashing*
- *Press the volume knob once. The WHITE status light will flash once. Headunit number 1 is now programmed. Repeat this step until the desired headunit number is reached - refer to the headunit legend*
- *When the desired headunit number is entered, press and hold the volume knob. The WHITE status light will illuminate. Release the volume knob. The RE_SWC will flash x times where x is the headunit number entered*

HEADUNIT LEGEND

HEADUNIT NUMBER	***HEADUNIT TYPE***
1	*RESISTANCE LEARNING*
2	*JVC*
3	*KENWOOD*

RESISTANCE LEARNING

Type 1

Resistance Learning Type 1 is used with headunits that support resitance learning and require momentary inputs when learning SWC functions

- *Open the Steering Wheel Control learning app/setting on your headunit*
- *Select the function you want to program on the headunit e.g. Volume+*
- *Perform the action with your volume knob e.g. for Volume+, rotate the volume knob clockwise one step*
- *The headunit will confirm that the function has been programmed*
- *Repeat for all required functions*
    - *Volume+/-*
    - *Button Short Press*
    - *Button Long Press*

RESISTANCE LEARNING
Type 2

Resistance Learning Type 2 is used with headunits that support resitance learning and require inputs to be held when learning SWC functions

- *Enable learning mode on RE_SWC controller by double-pressing the volume knob*
    - *The RE_SWC will confirm learning mode is enabled by rapidly flashing the WHITE status light*
- *Select the function you want to program on the headunit e.g. Volume+*
- *Perform the action with your volume knob e.g. for Volume+, rotate the volume knob clockwise one step*
- *The WHITE status light will illuminate for 4 seconds while the output is held.*

IMPORTANT: *Do not touch the SWC or headunit during this time to allow the headunit to configure itself*

- *Once the function is programmed, the WHITE status light will be turned off*
- *Repeat for all required functions*
    - *Volume+/-*
    - *Button Short Press*
    - *Button Long Press*

JVC FUNCTIONS

The following functions have been assigned to the RE_SWC for JVC headunits:

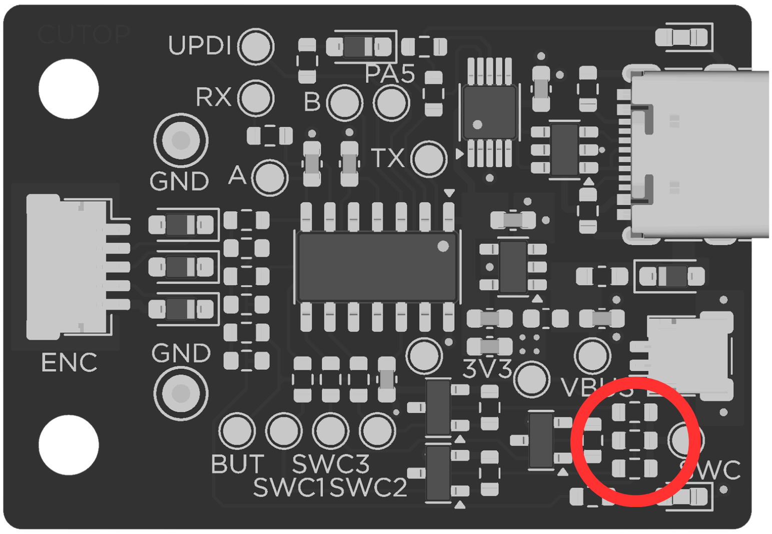
- *Volume +/-*
- *Button short press - Mute*
- *Button long press - Previous Track*
- *Button double press - Next Track*

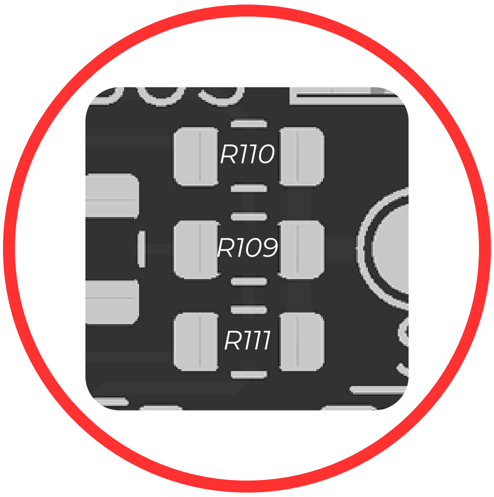
KENWOOD FUNCTIONS

The following functions have been assigned to the RE_SWC for Kenwood headunits:

- *Volume +/-*
- *Button short press - Play/Pause*
- *Button long press - Mute*
- *Button double press - Next track*

SWC OUTPUT RESISTORS





RESISTOR ID	***VALUE***	***OUTPUT***
R110	***1kΩ ± 1%***	***SWC OUT 2***
R109	***330Ω ± 1%***	***SWC OUT 1***
R111	***5.1kΩ ± 1%***	***SWC OUT 3***

RE_SWC SPECS

Input
5V USB-C
300mA AUTO RESETTING FUSE

Output
SWC OUT 1: 330Ω ± 1%
SWC OUT 2: 1kΩ ± 1%
SWC OUT 3: 5.1kΩ ± 1%

CONNECTORS
VOLUME KNOB: JST SHR-05V-S-B
SWC OUT: JST SHR-02V-S-B

MCU:
MICROCHIP ATTINY414
AVR 1 Series

Document Revision History

- *Rev 1 - Initial release*
- *Rev 2 - FW V2.0.0*
    - *Added instructions for JVC headunit interface*
    - *Added instructions for Kenwood headunit interface*